

数列专题习题册(中英双语版)

Sequence Special Topic Exercise Book (Chinese-English Bilingual Version)

一、选择题 I. Multiple Choice Questions

1. $\sqrt{5}$ 和 $\sqrt{5}$ 的等差中项是()

What is the arithmetic mean of $\sqrt{5}$ and $\sqrt{5}$? ()

A. $\sqrt{5}$

B. 2

C. $\pm\sqrt{5}$

D. 3

2. 若 $\{a_n\}$ 是等差数列, $a_1 = 4$, $a_3 = 12$, 则 $S_2 =$ ()

If $\{a_n\}$ is an arithmetic sequence, $a_1 = 4$, $a_3 = 8$, then $S_2 =$ ()

A. 6

B. 12

C. 18

D. 24

3. 4 和 16 的等比中项是()

What is the geometric mean of 4 and 16? ()

A. 20

B. 8

C. ± 8

D. 10

4. 已知 $\{a_n\}$ 为等比数列, $a_2 = 6$, $a_3 = 12$, 则公比 $q =$ ()

Given that $\{a_n\}$ is a geometric sequence, $a_2 = 6$, $a_3 = 12$, then the common ratio $q = ()$

- A. 4
- B. -3
- C. 3
- D. 2

5. 已知 $\{a_n\}$ 是等差数列, $a_1 = 3$, 公差 $d = 2$, 则 $a_5 = ()$

Given that $\{a_n\}$ is an arithmetic sequence, $a_1 = 3$, common difference $d = 2$, then $a_5 = ()$

- A. 11
- B. 7
- C. 9
- D. 13

6. 等差数列中, $a_3 = 7$, $S_3 = 15$, 则公差 $d = ()$

In an arithmetic sequence, $a_3 = 7$, $S_3 = 15$, then the common difference $d = ()$

- A. 2
- B. $\frac{4}{3}$
- C. 3
- D. 4

7. 已知等差数列 $\{a_n\}$ 的第三项是 6, 第七项是 22, 则公差 d 为()

Given that the 3rd term of the arithmetic sequence $\{a_n\}$ is 6 and the 7th

term is 22, then the common difference d is ()

A. 3

B. 4

C. 6

D. 5

8. 若递增等比数列 $\{a_n\}$ 中, $a_1 = 3$, $S_3 = 39$, 则公比 $q =$ ()

If in the increasing geometric sequence $\{a_n\}$, $a_1 = 3$, $S_3 = 39$, then the common ratio $q =$ ()

A. 3

B. 4

C. 2

D. 1

9. 下列数列是等差数列的是()

Which of the following sequences is an arithmetic sequence? ()

A. $1, 1 + \sqrt{2}, 1 + \sqrt{3}$ B. 1, 4, 16

C. $\lg 3, \lg 9, \lg 27$ D. -4, 1, 6

10. 若 $\{a_n\}$ 为等比数列, $a_1 = 4$, 公比 $q = \frac{1}{3}$, 则 $a_3 =$ ()

If $\{a_n\}$ is a geometric sequence, $a_1 = 4$, common ratio $q = \frac{1}{3}$, then $a_3 =$

()

A. $-\frac{4}{9}$ B. $\frac{4}{9}$

C. $-\frac{9}{4}$ D. $\frac{9}{4}$

二、填空题 II. Fill-in-the-Blank Questions

1. 数列 $\{a_n\}$ 的通项公式是 $a_n = \frac{a_{n-1}}{2^{n-1}} (n \geq 2)$, $a_2 = 4$, 则 $a_3 =$ _____。

For the sequence $\{a_n\}$, the general term formula is $a_n = \frac{a_{n-1}}{2^{n-1}} (n \geq 2)$, $a_2 = 4$, then $a_3 =$ _____.

2. 数列 $\{a_n\}$ 的通项公式是 $a_n = (-3)^2 \cdot \frac{1}{n}$, 则 $a_2 =$ _____。

For the sequence $\{a_n\}$, the general term formula is $a_n = (-3)^2 \cdot \frac{1}{n}$, then $a_2 =$ _____.

3. 已知 $a_1 = \frac{1}{3}$, $a_n = 3a_{n-1} + 2 (n > 1)$, 则 $a_2 =$ _____; $a_3 =$ _____。

Given $a_1 = \frac{1}{3}$, $a_n = 3a_{n-1} + 2 (n > 1)$, then $a_2 =$ _____; $a_3 =$ _____.

4. 数列 $\{a_n\}$ 的前 n 项和公式是 $S_n = 3n - 5$, 那么该数列第三项 $a_3 =$ _____。

The formula for the sum of the first n terms of the sequence $\{a_n\}$ is $S_n = 3n - 5$, then the 3rd term of the sequence $a_3 =$ _____.

5. 数列 $\{a_n\}$ 的通项公式是 $a_n = n^3 + 6$, 那么该数列首项 $a_1 =$ _____; 第二项 $a_2 =$ _____; 第三项 $a_3 =$ _____; 前三项和 $S_3 =$ _____。

For the sequence $\{a_n\}$, the general term formula is $a_n = n^3 + 6$, then the first term of the sequence $a_1 =$ _____; the second term $a_2 =$ _____; the third term $a_3 =$ _____; the sum of the first three terms $S_3 =$ _____.

6. 11 和 13 的等差中项是 _____。

The arithmetic mean of 11 and 13 is _____.

7. 等差数列中, $a_1 = 4$, $d = -4$, 则 $a_7 =$ _____。

In an arithmetic sequence, $a_1 = 4$, $d = -4$, then $a_7 =$ _____.

8. 等差数列中, $a_1 = 2$, $a_7 = 20$, 则公差 $d =$ _____。

In an arithmetic sequence, $a_1 = 2$, $a_7 = 20$, then the common difference $d =$ _____.

9. 等差数列中, $a_1 = -3$, $a_5 = 11$, 则公差 $d =$ _____; $a_3 =$ _____。

In an arithmetic sequence, $a_1 = -3$, $a_5 = 11$, then the common difference $d =$ _____; $a_3 =$ _____.

10. 等差数列中, $a_7 = -11$, $d = -2$, 则 $a_1 =$ _____。

In an arithmetic sequence, $a_7 = -11$, $d = -2$, then $a_1 =$ _____.

11. 等比数列中, $a_1 = 4$, $q = -3$, 则 $S_3 =$ _____。

In a geometric sequence, $a_1 = 4$, $q = -3$, then $S_3 =$ _____.

12. 等比数列中, $a_3 = 5$, $a_4 = 10$, 那么公比 $q =$ _____。

In a geometric sequence, $a_3 = 5$, $a_4 = 10$, then the common ratio $q =$ _____.

13. 等比数列中, $a_2 = 10$, $a_4 = 40$, 那么公比 $q =$ _____; $a_1 =$ _____。

In a geometric sequence, $a_2 = 10$, $a_4 = 40$, then the common ratio $q =$ _____; $a_1 =$ _____.

14. 等比数列中, $q = \frac{1}{3}$, $a_3 = \frac{1}{9}$, 那么 $a_1 =$ _____。

In a geometric sequence, $q = \frac{1}{3}$, $a_3 = \frac{1}{9}$, then $a_1 =$ _____.

15. $\frac{\sqrt{6}}{3}$ 和 $\frac{\sqrt{6}}{3}$ 的等比中项是 _____。

The geometric mean of $\frac{\sqrt{6}}{3}$ and $\frac{\sqrt{6}}{3}$ is _____.

16. 3 和 12 的等比中项是 _____。

The geometric mean of 3 and 12 is _____.

17. 等差数列中, $a_8 = 11$, $d = 3$, 那么 $a_1 =$ _____; $S_8 =$ _____。

In an arithmetic sequence, $a_8 = 11$, $d = 3$, then $a_1 =$ _____; $S_8 =$ _____.

18. 等差数列中, $a_1 = -6$, $d = \frac{1}{3}$, 则 $a_7 =$ _____。

In an arithmetic sequence, $a_1 = -6$, $d = \frac{1}{3}$, then $a_7 =$ _____.

19. 等差数列中, $a_3 = 5$, $a_8 = 15$, 则 $a_1 =$ _____。

In an arithmetic sequence, $a_3 = 5$, $a_8 = 15$, then $a_1 =$ _____.

20. 等差数列中, $a_1 = 2$, $a_4 = 8$, 则 $S_4 =$ _____。

In an arithmetic sequence, $a_1 = 2$, $a_4 = 8$, then $S_4 =$ _____.

21. 等差数列中, $a_2 = 3$, $a_6 = 11$, 则 $a_{10} =$ _____。

In an arithmetic sequence, $a_2 = 3$, $a_6 = 11$, then $a_{10} =$ _____.

22. 等比数列中, $a_3 = 5$, $a_4 = 15$, 那么公比 $q =$ _____。

In a geometric sequence, $a_3 = 5$, $a_4 = 15$, then the common ratio $q =$ _____.

等比数列中, $a_1 = \frac{1}{3}$, $a_4 = 9$, 那么公比 $q =$ _____。

23. In a geometric sequence, $a_1 = \frac{1}{3}$, $a_4 = 9$, then the common ratio $q =$ _____.

24. 等比数列中, $a_6 = 6$, $a_5 = 3$, 那么公比 $q =$ _____。

In a geometric sequence, $a_6 = 6$, $a_5 = 3$, then the common ratio $q =$ _____.

25. 等比数列中, $a_1 = 2$, $q = 5$, 则 $S_3 =$ _____。

In a geometric sequence, $a_1 = 2$, $q = 5$, then $S_3 =$ _____.

26. 等比数列中, $a_3 = 8$, $a_5 = 32$, 那么公比 $q =$ _____; $a_1 =$ _____。

In a geometric sequence, $a_3 = 8$, $a_5 = 32$, then the common ratio $q =$ _____; $a_1 =$ _____.

27. 7 和 9 的等差中项是 _____。

The arithmetic mean of 7 and 9 is _____.

28. $\sqrt{7}$ 和 $4\sqrt{7}$ 的等差中项是 _____。

The arithmetic mean of $\sqrt{7}$ and $4\sqrt{7}$ is _____.

29. $2a + b$ 和 $2a - b$ 的等差中项是 _____。

The arithmetic mean of $2a + b$ and $2a - b$ is _____.

30. 5 和 15 的等比中项是 _____。

The geometric mean of 5 and 15 is _____.

31. $\frac{\sqrt{8}}{2}$ 和 $\frac{\sqrt{8}}{2}$ 的等比中项是 _____。

The geometric mean of $\frac{\sqrt{8}}{2}$ and $\frac{\sqrt{8}}{2}$ is _____.

32. 3 和 15 的等差中项是 _____; 等比中项是 _____。

The arithmetic mean of 3 and 15 is _____; the geometric mean is _____.

33. 数列的通项公式为 $a_n = n^2 - 2$, 则它的第三项 $a_3 =$ _____。

The general term formula of the sequence is $a_n = n^2 - 2$, then its 3rd term $a_3 =$ _____.

34. 数列的前 n 项和公式为 $S_n = 6(n + 3)$, 则它的第二项 $a_2 =$ _____。

The formula for the sum of the first n terms of the sequence is $S_n = 6(n + 3)$, then its second term $a_2 =$ _____.

三、计算题 III. Calculation Questions

1. 等差数列中, $a_3 = 9$, $a_7 = 21$, 求首项 a_1 和前 4 项和 S_4 。

In an arithmetic sequence, $a_3 = 9$, $a_7 = 21$, find the first term a_1 and the sum of the first 4 terms S_4 .

2. 等差数列中, $a_2 = 6$, $S_5 = 40$, 求首项 a_1 和公差 d 。

In an arithmetic sequence, $a_2 = 6$, $S_5 = 40$, find the first term a_1 and the common difference d .

3. 等差数列中, $a_3 = 7$, $a_6 = 25$, 求:

In an arithmetic sequence, $a_3 = 7$, $a_6 = 25$, find:

(1) 首项 a_1 和公差 d 的值; The values of the first term a_1 and the common difference d ;

(2) a_7 的值; The value of a_7 ;

(3) 前 4 项和 S_4 的值。The value of the sum of the first 4 terms S_4 .

4. 等比数列 $\{a_n\}$ 中, $a_4 = 3$, $a_7 = 24$, 求:

In the geometric sequence $\{a_n\}$, $a_4 = 3$, $a_7 = 24$, find:

(1) 首项 a_1 和公比 q ; The first term a_1 and the common ratio q ;

(2) 前 5 项和 S_5 。The sum of the first 5 terms S_5 .

5. 在等差数列 $\{a_n\}$ 中, $a_1 = 9$, $a_3 = 3$, 求:

In the arithmetic sequence $\{a_n\}$, $a_1 = 9$, $a_3 = 3$, find:

(1) 9 和 3 的等差中项 A ; The arithmetic mean A of 9 and 3;

(2) 公差 d ; The common difference d ;

(3) 前四项和 S_4 。The sum of the first four terms S_4 .